

Quality of life predictors in patients with chronic inflammatory demyelinating polyradiculoneuropathy

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Abstract Chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is a chronic disease which can lead to many functional impairments, and like most other chronic disorders it might significantly affect quality of life (QoL). Information about QoL in patients with CIDP from developing countries is still lacking. We, therefore, sought to complete these data mosaic by investigating QoL in patients with CIDP from Serbia and surrounding countries. Our study comprised 106 patients diagnosed with CIDP. QoL was investigated using the Serbian version of the SF-36 questionnaire. The Medical Research Council 0–5 point scale, INCAT motor and sensory scores, Krupp's Fatigue Severity Scale, and Beck Depression Inventory were also used. Factors that significantly correlated with SF-36 total score in univariate analysis were included in the multiple linear regression analysis. Physical domains of the SF-36 were more affected than mental, and the overall score was 56.6 ± 25.4 . Significant predictors of worse SF-36 score in our patients with CIDP were severe fatigue ($\beta = -0.331$, $p < 0.01$), higher INCAT motor score ($\beta = -0.301$,

$p < 0.01$), depression ($\beta = -0.281$, $p < 0.01$), being unemployed/retired ($\beta = -0.188$, $p < 0.05$), and shorter duration of CIDP ($\beta = +0.133$, $p < 0.01$). QoL was reduced in CIDP patients, especially in physical domains. Patients with presence of fatigue and depression, with more severe motor disability, unemployed/retired ones, and those with shorter duration of the disease need special attention of clinicians since they could be at higher risk to have worse QoL.

Keywords Chronic inflammatory demyelinating polyradiculoneuropathy · Quality of life · SF-36 · Predictors · Fatigue · Depression

Introduction

Chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is an autoimmune neuropathy which is characterized by progressive, stepwise, or recurrent symmetric proximal and distal weakness, sensory dysfunction, and absent or reduced tendon reflexes of all extremities, developing over at least 2 months [1]. It is a chronic disease which can lead to many functional impairments of unpredictable duration and sequelae [2], thus like most other chronic disorders CIDP may significantly affect patients' quality of life (QoL).

Some previous investigations suggest that QoL in CIDP patients is affected only by physical disabilities and sensory symptoms, whereas their mental and emotional status is not disturbed [2]. On the other hand, some authors reported that different neuropsychiatric conditions could also affect QoL [3]. Different factors besides physical disability such as pain, fatigue, anxiety and depression, could affect not only the patients' QoL but even the course and outcome of CIDP [4, 5]. The main reason for these QoL variations reported so far could be patients' different social and cultural environment.

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Furthermore, most previous QoL studies included smaller cohorts of CIDP patients (from 4 to up to 59), and were mostly conducted in developed countries [3, 4, 6–8]. To our knowledge, Merkies et al., who primarily investigated the association between different impairments, activity, participation restrictions, and QoL using data from an ICE trial, had the largest cohort of patients ($n = 117$) [9]. Assessment of QoL in CIDP patients from developing countries are still lacking and we aimed to complete these data mosaic analyzing CIDP patients from Serbia, Bosnia and Herzegovina and Montenegro.

The aim of our study was to analyze QoL in a large cohort of prevalent CIDP patients using SF-36, as well as to identify sociodemographic, clinical and psychological predictors influencing QoL in these patients.

Patients and method

In August 2016, the Inflammatory Neuropathy Study of Serbia (INeSS) was designed to collect as many as patients with CIDP from Serbia, Republic of Srpska (Bosnia and Herzegovina), and Montenegro. We tried to reach all patients diagnosed with CIDP in tertiary health care centers in a 10-year period between 2007 and 2016. Only patients who fulfilled EFNS/PNS diagnostic criteria were included (EFNS/PNS Guidelines) [1]. CIDP variant was also diagnosed according to the EFNS/PNS criteria. Among 190 patients identified in this way, at the time of testing 26 were deceased, 32 were lost to follow-up, and 13 declined to participate in the study. Since we were not able to analyze sera for the presence of anti-MAG antibodies, all patients positive for IgM paraprotein were excluded to have a CIDP-only group. In addition, nine patients were excluded due to other severe diseases (stroke, hematological malignancy, connective tissue disease, motor neuron disease, and neurofibromatosis), making the overall number of 106 patients tested. The study was approved by the Ethical Board of the Neurology Clinic, Clinical Center of Serbia, and all patients gave informed consent to participate in the study.

We collected sociodemographic data including gender, age, education, profession, and duration of the disease from patients' medical records at time of diagnosis and at time of testing. Therapeutic modality and response to treatment were also included, as well as presence of significant conditions such as diabetes mellitus and monoclonal gammopathy of undetermined significance (MGUS), except IgM since these patients were excluded. The Medical Research Council (MRC) 0–5 point scale (0—without movement, 5—normal strength) was used to evaluate the muscle strength [10]. Degree of functional disability was measured using the INCAT motor score (Inflammatory Neuropathy Cause and

Treatment score) [11]. The objective sensory deficits of our patients were rated using the INCAT sensory score [12].

Krupp's Fatigue Severity Scale (FSS) was used for the evaluation of severity of fatigue. It is a nine-item questionnaire where a score above 36 means significant fatigue [13]. Level of depression was established using the Beck Depression Inventory (BDI). Depression was considered if the score was ≥ 11 [14].

As a measure of health-related QoL, each patient filled out the Serbian version of the SF-36 questionnaire [15], which is a generic measure that combines eight general health concepts: physical functioning (PF), role physical (RP), bodily pain (BP), general health (GH), vitality (VT), social functioning (SF), role emotional (RE), and mental health (MH). Besides the total SF-36 score, physical composite score (PCS) and mental composite score (MCS) are two main scores to summarize these eight scales. All scores are interpreted with a 0–100 scale, where higher numbers represent better QoL.

Normality of data was tested by the Kolmogorov–Smirnov test. For group comparisons, χ^2 test, Mann–Whitney U test and Student t test were used, as appropriate. Correlations were assessed using Spearman's rho. Factors that significantly correlated with SF-36 total score ($p < 0.05$) in univariate analysis were included in the multiple linear regression analysis (stepwise method). Stepwise criteria were as follow: probability of F to enter variable was ≤ 0.05 , and probability to remove variable ≥ 0.10 . SPSS software created five models with SF-36 total score as a dependent variable. Model with the highest adjusted R^2 value was taken into account. In this way, factors significantly associated with QoL were identified. For all statistical tests, significant testing was two-sided, with alpha set at 0.05 for statistical significance and 0.01 for high statistical significance.

Results

Main sociodemographic and clinical data of investigated patients are shown in Table 1. Clinically typical variant of CIDP was present in 60% of patients. Majority (86%) of our 106 patients fulfilled the definite EFNS/PNS electrophysiological criteria, while 1% fulfilled the probable and 13% possible criteria. Outcome of the disease at the moment of testing is presented in Fig. 1. MRC score and INCAT (motor, sensory and total) showed improvement compared to baseline data. In our cohort, 43% of patients complained of severe fatigue, and mean FSS score was 34.8 ± 17.2 . Depression appeared in 30% of our patients, and mean score on BDI scale was 8.2 ± 7.6 .

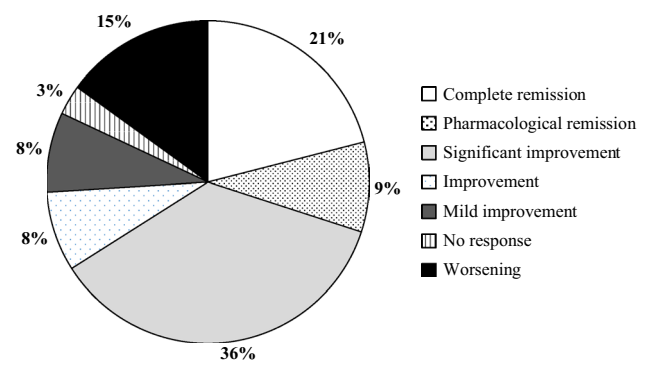
Results on the SF-36 questionnaire are shown in Table 2. RP was the health concept with the lowest score, while SF, MH and BP were domains with the highest scores. Influence

Table 1 Sociodemographic and clinical features of investigated CIDP patients ($n = 106$) at time of diagnosis and at time of testing

CIDP features	At diagnosis	At testing
Male gender	63%	63%
Age (years)	53.5 ± 15.1	60.3 ± 13.2
Education (years)	11.0 ± 2.7	12.2 ± 2.8
Profession	–	
Student		1.8%
Employed		14.2%
Unemployed		16.0%
Retired		65.1%
Unknown		2.8%
Disease duration (years)	0.9 ± 0.5	6.8 ± 6.4
INCAT sensory		
Arms	2.3 ± 2.2	1.4 ± 1.9
Legs	4.0 ± 2.1	2.8 ± 2.1
Total	6.3 ± 3.7	4.2 ± 3.6
INCAT motor		
Arms	1.5 ± 1.2	0.6 ± 0.9
Legs	1.9 ± 1.4	1.1 ± 1.3
Total	3.4 ± 2.1	1.8 ± 2.0
MRC sum score	48.4 ± 9.6	53.9 ± 9.7
FSS	–	34.8 ± 17.2
Fatigue (%)		43.4%
BDI	–	8.2 ± 7.6
Depression		30.2%
CIDP variants	–	
Typical		60.4%
Sensory ataxic		9.4%
Motor		9.4%
Paraparetic		9.4%
DADS		8.5%
LSS		3.8%
Comorbid disorders		
Diabetes mellitus	17.9%	19.8%
MGUS	14.2%	16.0%
Therapy (any time)		–
Oral prednisone	85.8%	
Pulsed methylprednisolone	18.9%	
IVIg	38.7%	
TPE	3.8%	
Immunosuppressant drugs	12.2%	

INCAT Inflammatory Neuropathy Cause and Treatment, MRC Medical Research Council sum score, FSS Fatigue Severity Scale, BDI Beck Depression Inventory, LSS Lewis–Sumner syndrome, DADS distal acquired demyelinating symmetric neuropathy, MGUS monoclonal gammopathy of undetermined significance, IVIg intravenous immunoglobulin, TPE therapeutic plasma exchange

of different sociodemographic and clinical features on SF-36 total scores are presented in Table 3. It is of note that neither of the patients' features at time of diagnosis

**Fig. 1** Outcome of the disease in our cohort of patients ($n = 106$)**Table 2** Results on the SF-36 questionnaire in CIDP patients ($n = 106$)

SF-36 scale	Score
Physical functioning (SF)	52.1 ± 33.6
Role physical (RF)	42.9 ± 42.8
Bodily pain (BP)	60.9 ± 30.6
General health (GH)	53.6 ± 23.3
Vitality (VT)	55.7 ± 25.4
Social functioning (SF)	70.4 ± 27.2
Role emotional (RE)	53.5 ± 46.2
Mental health (MH)	63.4 ± 20.6
Physical composite score (PCS)	53.1 ± 26.4
Mental composite score (MCS)	59.3 ± 24.0
Total SF-36 score	56.6 ± 25.4

correlated with SF-36 total score. In addition, we did not observe that any therapeutic modality correlated with SF-36, except for pulsed methylprednisolone therapy—patients that received pulsed therapy had total SF-36 score 43.3 ± 23.9 compared to 60.2 ± 24.7 in those who did not receive this therapy ($p < 0.01$). Using the multiple linear regression analysis we established that the significant predictors of worse SF-36 score in our patients with CIDP were severe fatigue ($\beta = -0.331$, $p < 0.01$), higher INCAT motor score ($\beta = -0.301$, $p < 0.01$), depression ($\beta = -0.281$, $p < 0.01$), being unemployed/retired ($\beta = -0.188$, $p < 0.05$) and shorter duration of CIDP ($\beta = +0.133$, $p < 0.01$) (Table 4).

Discussion

We conducted this research to fill in the gap in knowledge about QoL in CIDP patients from developing countries. QoL was reduced in our patients, especially in physical domains, and it was influenced not only by their physical status but also by psychological and social factors.

Table 3 Association between different sociodemographic/clinical features and total SF-36 score ($n = 106$)

CIDP features	Association with SF-36 total score
Gender	
Male	57.6 ± 24.5
Female	54.8 ± 27.1
Age*	$\rho = -0.245$
Education**	$\rho = -0.382$
Profession**	
Employed	81.9 ± 13.3
Unemployed	52.5 ± 28.9
Retired	50.5 ± 22.3
Disease duration**	$\rho = +0.306$
INCAT sensory	
Arms**	$\rho = -0.388$
Legs**	$\rho = -0.389$
Total**	$\rho = -0.439$
INCAT motor	
Arms**	$\rho = -0.559$
Legs**	$\rho = -0.583$
Total**	$\rho = -0.652$
MRC sum score**	$\rho = +0.615$
FSS**	$\rho = -0.724$
BDI**	$\rho = -0.662$
CIDP variants	
Typical	55.6 ± 24.4
Atypical	58.1 ± 27.0
Diabetes mellitus*	
Yes	60.4 ± 24.8
No	47.9 ± 24.1
Outcome**	
Remission, significant improvement	66.0 ± 24.7
Improvement	57.3 ± 23.2
No improvement	34.8 ± 20.1

INCAT Inflammatory Neuropathy Cause and Treatment, MRC Medical Research Council sum score, FSS Fatigue Severity Scale, BDI Beck Depression Inventory

* $p < 0.05$, ** $p < 0.01$

In a large cohort of our CIDP patients, physical scores of the SF-36 were more impaired than mental ones, which is in accordance with previous studies on CIDP [7, 16, 17]. Our results were also very similar to a previous Swedish study, except that RE domain score was lower in our group [7]. Although the Swedish cohort ($n = 21$) was notably smaller than ours ($n = 106$), the main reason for this incompatibility in RE scores could be the difference in the health care systems of our countries. The affirmation of our findings can also be found in an insufficient number of papers in other inflammatory neuropathies such as multifocal motor neuropathy, where physical domains were also more impaired

Table 4 Significant predictors of worse SF-36 total score in CIDP patients ($n = 106$)—multiple linear regression analysis (stepwise method)

Feature	β
Fatigue	-0.331^{**}
Higher INCAT motor	-0.301^{**}
Depression	-0.281^{**}
Being unemployed/retired	-0.188^{*}
Duration of disease	$+0.133^{*}$
R^2 adjusted	0.691

INCAT Inflammatory Neuropathy Cause and Treatment

* $p < 0.05$, ** $p < 0.01$

than mental ones and individual SF-36 scores were similar as in our CIDP group [18–20]. Moreover, SF-36 scores of mixed CIDP/MGUS polyneuropathy cohort from the Netherlands were very similar to our findings [21]. More severe impairment in physical domains of QoL was also observed in patients with other neuromuscular disorders [22–24]. On the other hand, both physical and mental domains are equally impaired in myotonic dystrophies, but these are disorders that affect the central nervous system [25, 26]. However, CIDP usually does not affect the brain, and we suppose that these patients are able to cope better with their illness and to preserve mental health domains.

Results of our study presented worse INCAT motor score as a significant predictor of the worse SF-36 score, indicating that physical disability can cause an important disturbance in patients' daily activities. However, physical disability was not the only, and not even the most significant predictor of QoL. In line with this, Merkies et al. showed correlation between physical disability and SF-36 scores, where the INCAT measure explained only 20% variance of PCS of SF-36 [27].

Considering that chronic diseases have a mostly negative impact on patients' mental health [28], we investigated presence of depression in our cohort of CIDP patients. The symptoms of at least mild depression appeared in 30% of our patients, which fits with previous data of 9–40% in CIDP cohorts [4, 29]. Although a bicycle training study proved that physical exercising improved the patients' mental status and reduced the level of depression and anxiety [30], the impact of physical exercising was more obvious on the PCS than on the MCS of the SF-36 [4]. Therefore, we recommend that patients with CIDP should also get psychiatric support, when necessary.

As it was proven in some earlier case series studies, fatigue can be a significant burden for CIDP patients and even a presenting symptom of the disease [31]. About 43% of our patients reported a presence of severe fatigue, which is in accordance with the previous results of up to 68% of

patients with severe fatigue, underlining fatigue was a serious symptom even in patients with residual neurological deficit [7, 32]. Earlier results showed an association between higher level of fatigue and lower SF-36 scores, where the strongest correlation was found in the physical domains [4]. Subsequently, our findings indicate that severe fatigue is one of the most significant predictors of the worse SF-36 score in CIDP patients and that it should be adequately treated to increase patients' QoL.

One of the independent predictors of the SF-36 score in our CIDP cohort was employment status. This influence could not be explained only by the fact that unemployed patients have more severe physical disability, but it is possible that being unemployed or retired is an additional psychological burden since these patients may feel useless. One recent UK study noted that 32% of CIDP patients retired early because of their disease [29]. We suppose that special attention should be paid to keep CIDP patients employed and to secure them professional reorientation when needed.

Although it is expected that longer duration of disease could have a negative impact on QoL due to a chronic disease burden, our results indicated that there was a statistically significant correlation between shorter duration of CIDP and worse SF-36 score. This could be explained by positive effects of therapy (around 60% of our patients had remission or significant improvement at the moment of testing), but also by possible coping strategies. Our findings emphasize that the first years after being diagnosed with CIDP could be the most demanding for patients, and clinicians should be aware of this.

It is of note that our patients that received pulsed methylprednisolone actually had worse QoL scores, although this was not an independent predictor according to the multiple linear regression analysis. Further analysis did not show that our patients treated with pulsed steroids had more severe disease, thus we believe that possible side effects or frequent re-hospitalizations for pulsed therapy might affect patients' QoL. A previous Italian study showed that a similarly high proportion of patients treated with IVIg or pulsed methylprednisolone eventually relapse after therapy discontinuation, but the median time to relapse was even longer after pulsed steroids [33]. Furthermore, 17% of patients withdrew or had serious adverse events with pulsed methylprednisolone compared with 9% who withdrew or died during or after IVIg, but this difference was not of statistical significance. Effect of these two therapeutic modalities on patients' QoL was not assessed. One Dutch study compared effects of pulsed oral dexamethasone vs. short-term prednisolone treatment [34]. Cure or long-term remission was achieved in about one-quarter of patients. Dexamethasone pulsed therapy seemed to be a more appropriate choice because it led to a faster improvement, slightly longer remissions, relatively fewer relapses, and less adverse events when compared to

prednisolone. However, effects on QoL were not analyzed in these two CIDP groups. Possible directions for further research are head-to-head comparisons of different steroid modalities (oral prednisone, pulsed intravenous methylprednisolone, and pulsed oral dexamethasone), analyzing their influence not only on objective outcome measures and side effects, but also on subjective perception of QoL.

Main limitation of our study is its cross sectional design, and further prospective data will be of great interest. In addition, analysis of QoL with a disease specific questionnaire might give deeper insights into the health status of CIDP patients. Another limitation of our study is that we did not include all potentially important factors that might affect QoL such as pain, anxiety, acceptance of illness, coping strategies, perceived social support, personality traits, etc. However, factors included in our model explained majority of the variance (69%) of the SF-36 total score.

Conclusion

QoL scores were reduced in CIDP patients, particularly in physical domains. Patients who are at high risk of having low QoL perception are those with presence of fatigue and depression, patients with more severe motor disability, unemployed/retired ones, and those with shorter duration (shorter treatment) of the disease.

Compliance with ethical standards

Conflicts of interest None of the authors has any conflict of interest to disclose.

Ethical standards All authors comply with Springer's ethical policies.

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